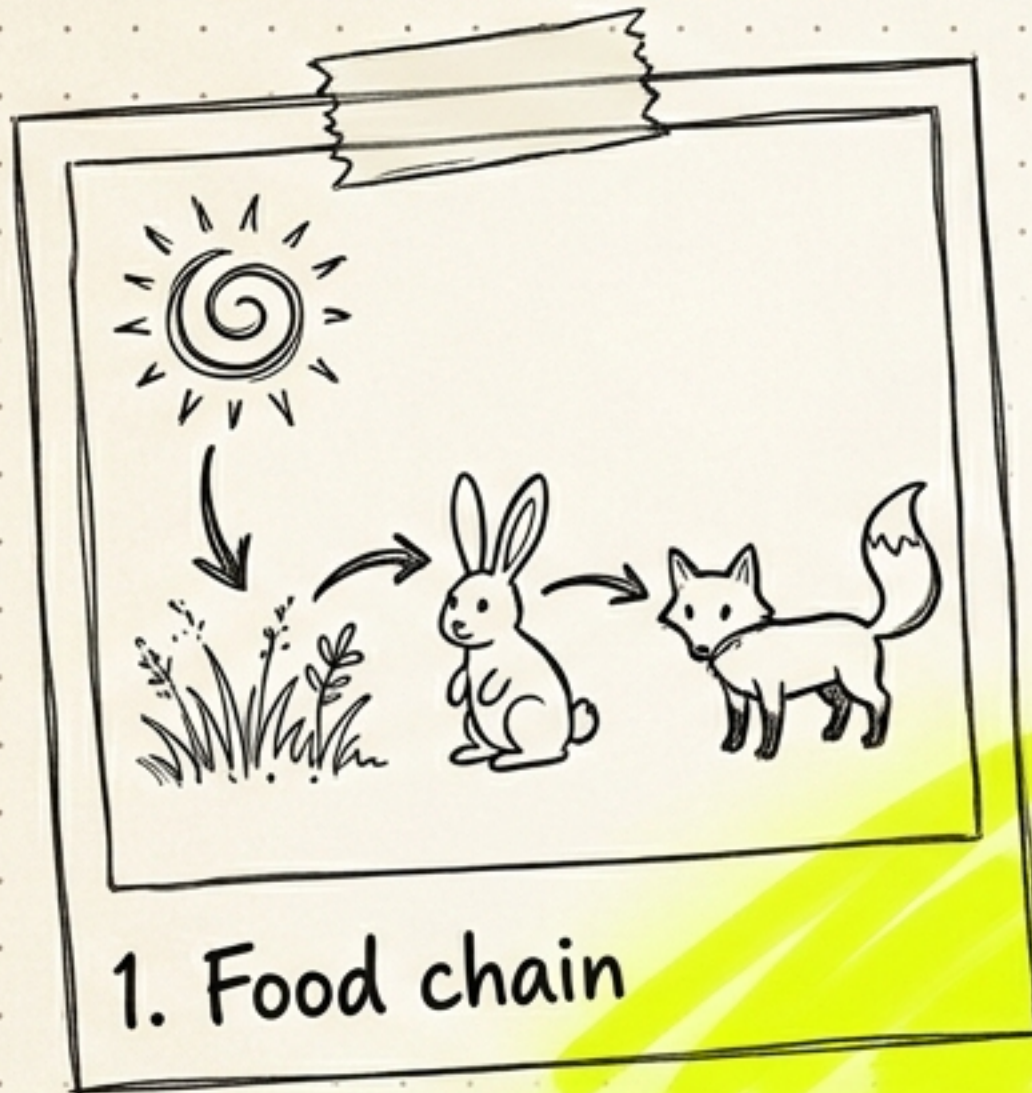


Photosynthesis

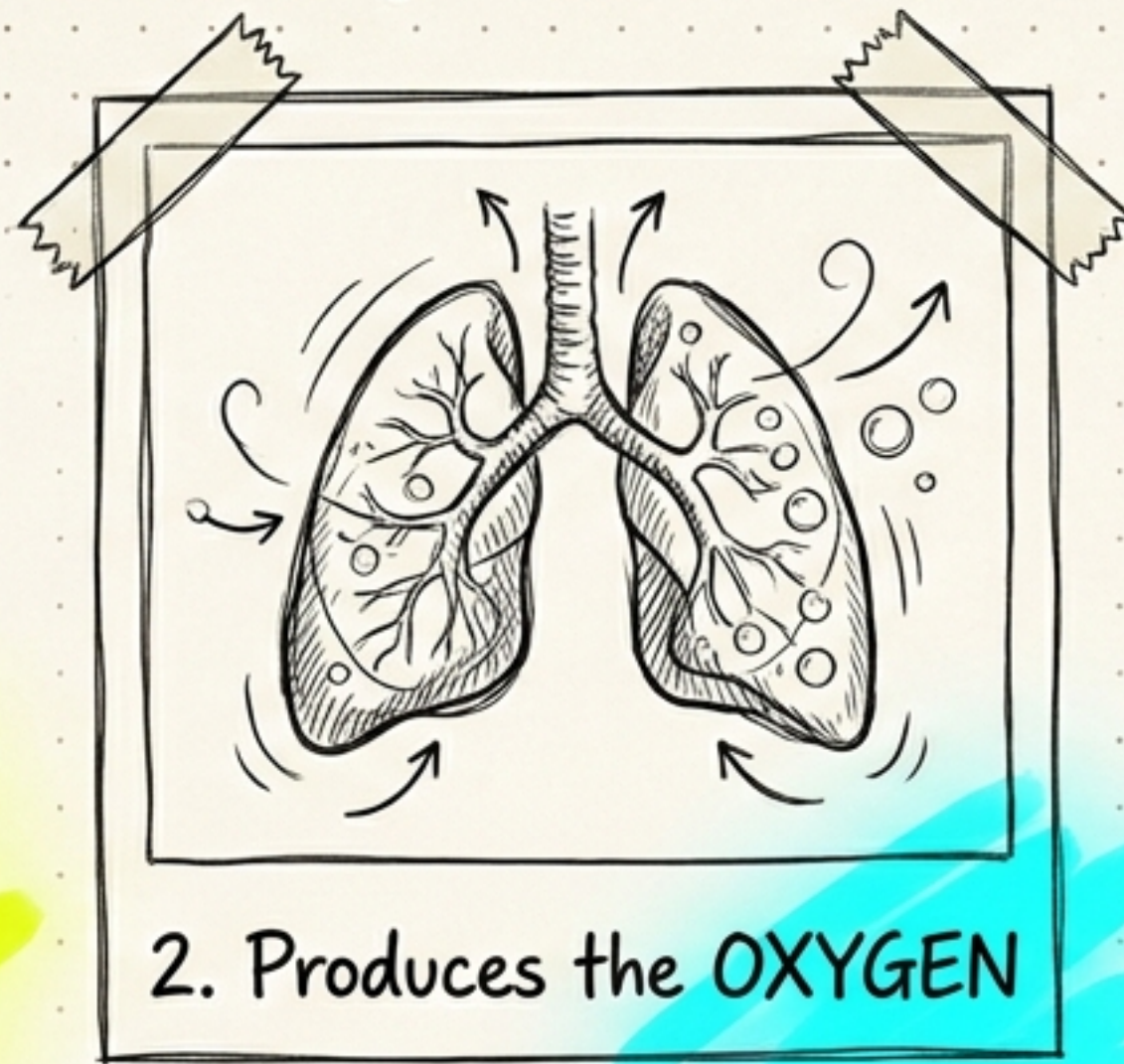
How Light Becomes Life.



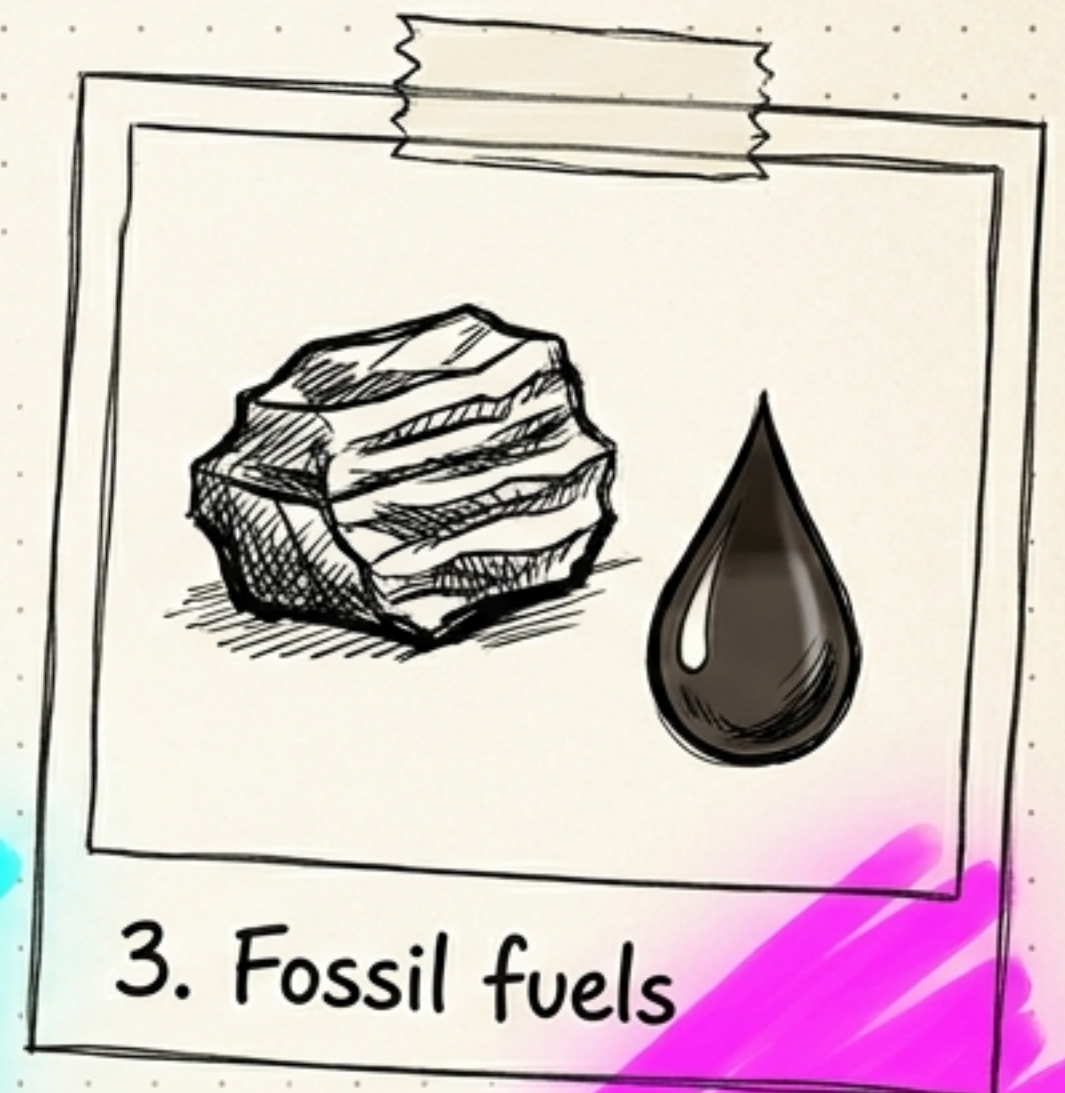
You would be dead without it. This 450-million-year-old process sustains the entire globe.



1. The base of almost all **FOOD CHAINS**. Plants synthesize organic matter from inorganic molecules.

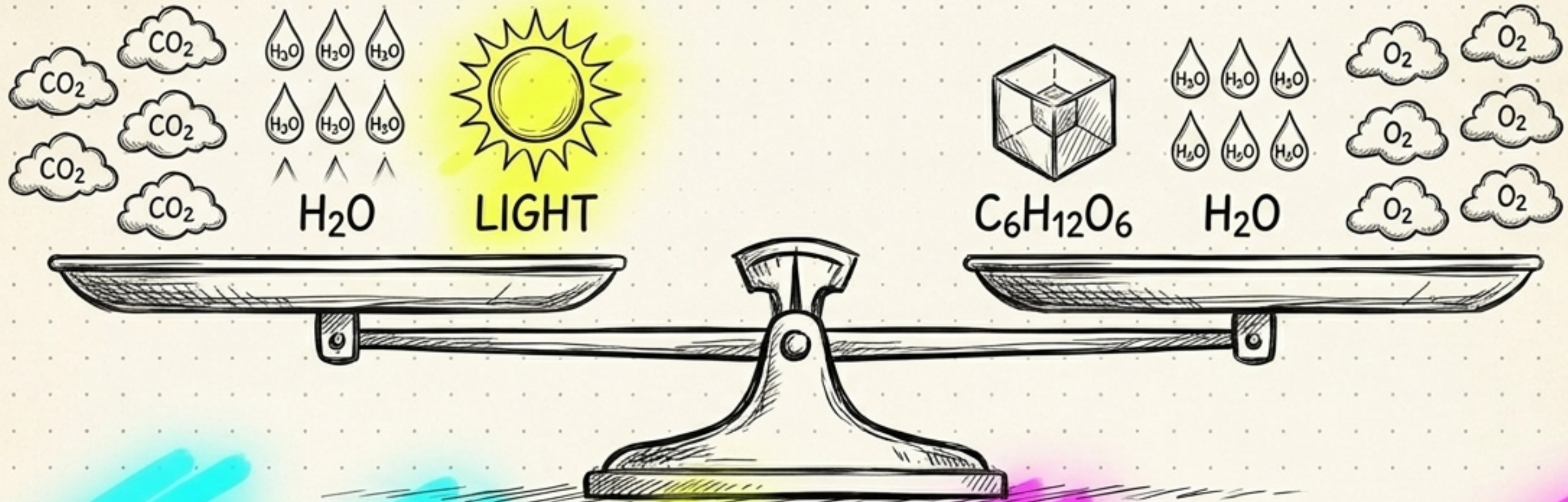


2. Produces the **OXYGEN** required for aerobic respiration.



3. Created the **FOSSIL FUELS** (coal, oil, gas) we use today from ancient plants.

The Master Recipe:

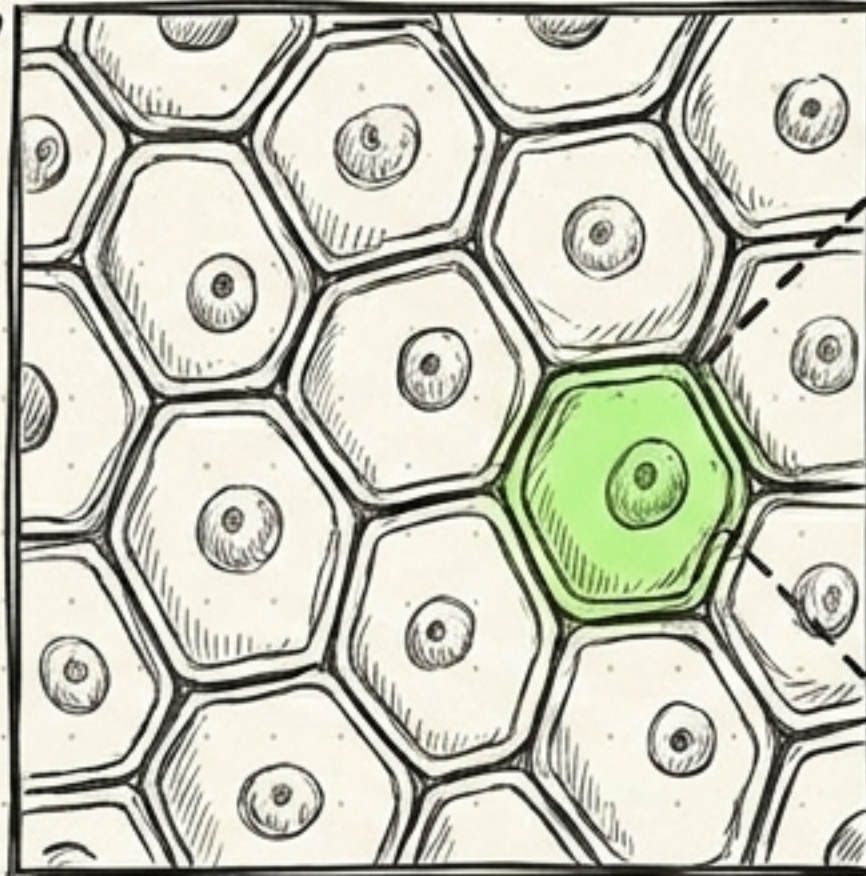


Crucial Note: The oxygen we breathe comes from the splitting of the water molecule, NOT the CO_2 !

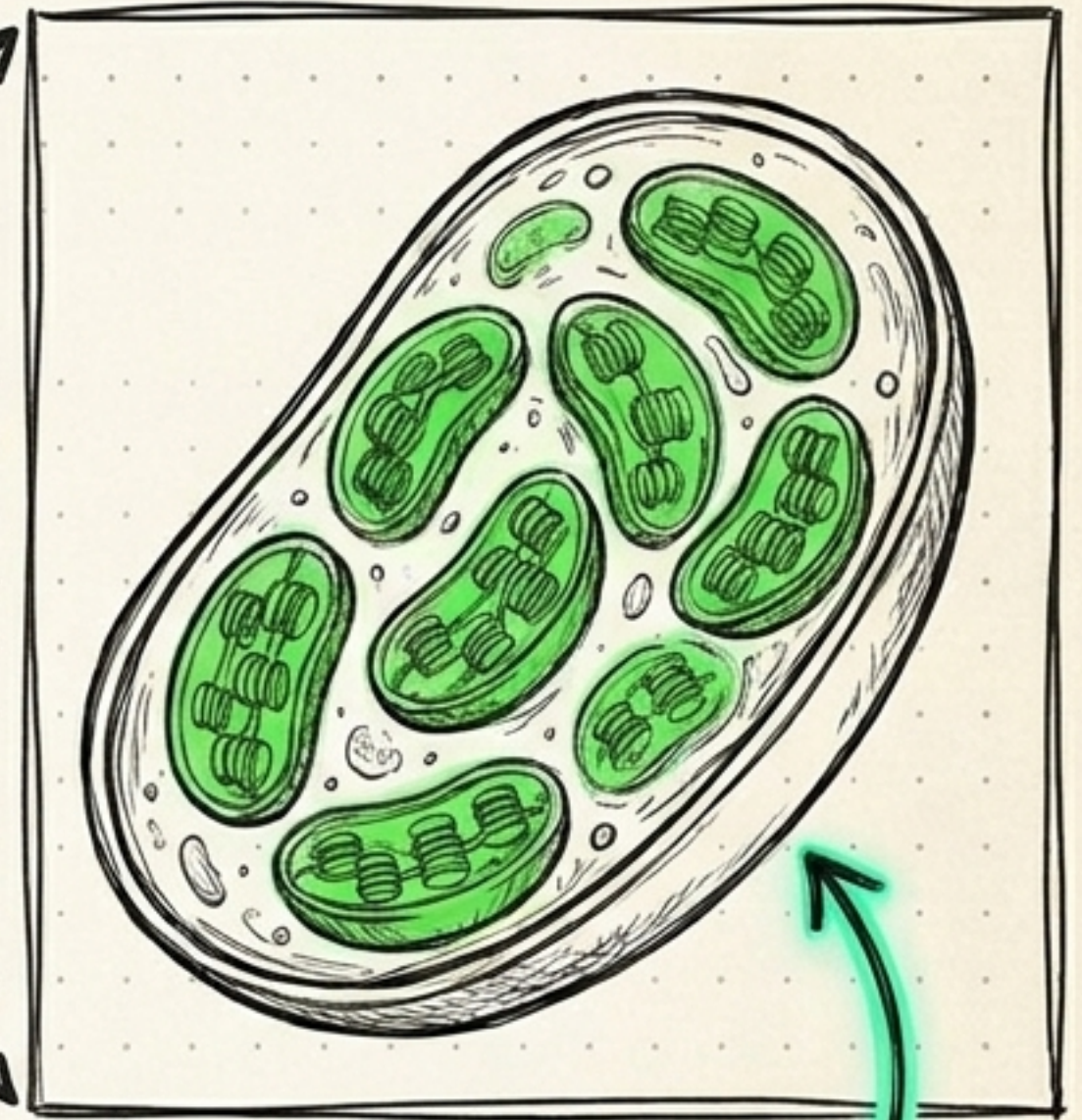
Where does it happen?
The green parts of the plant.



Leaf



Mesophyll Cells: Specialized leaf cells that act as the plant's solar factories.



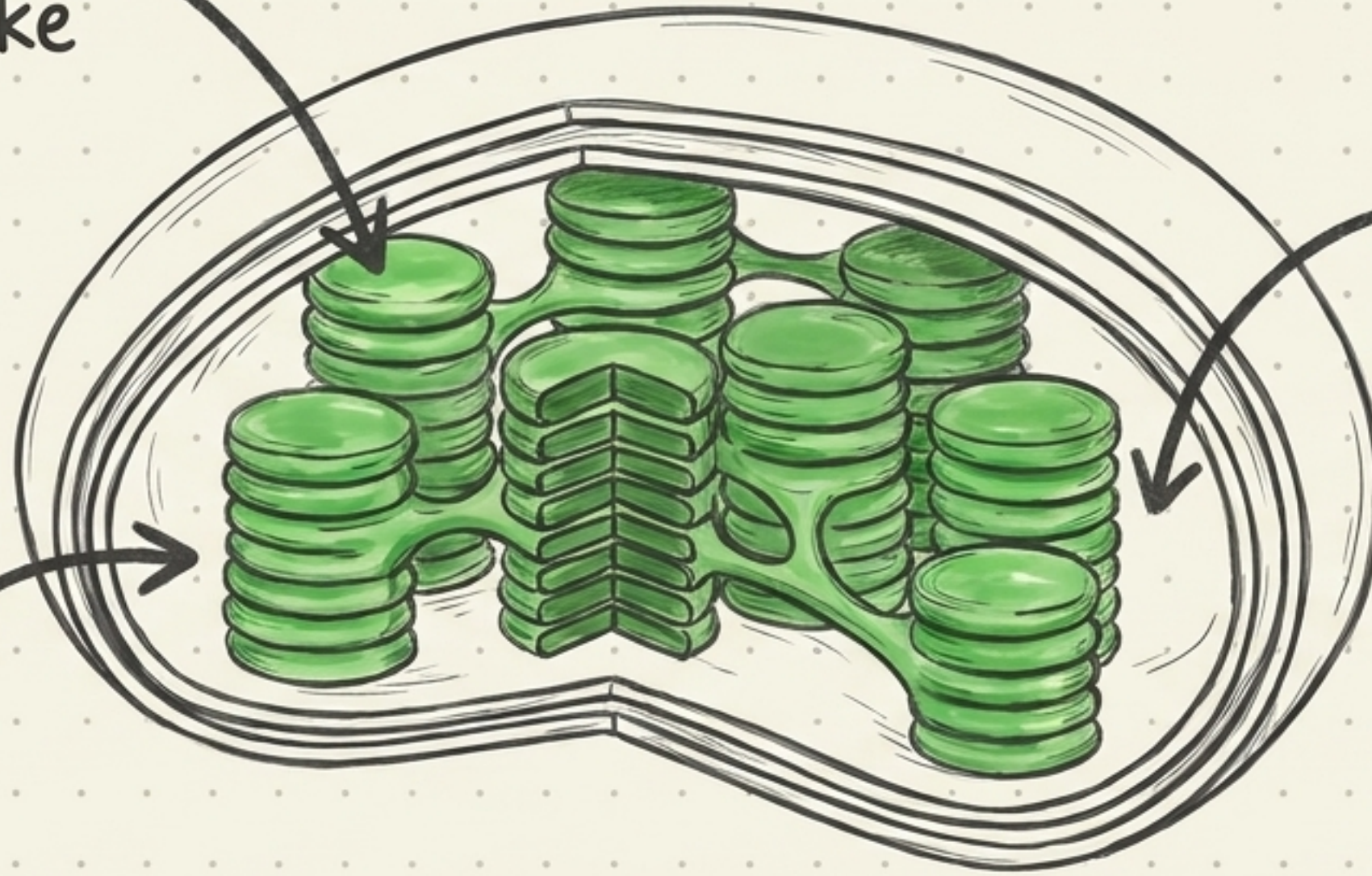
Chloroplasts: The actual pigment-containing organelles where the magic happens.

Thylakoids:

Flattened, disc-like membranes. This is where light is captured.

Grana:

Stacks of thylakoids.

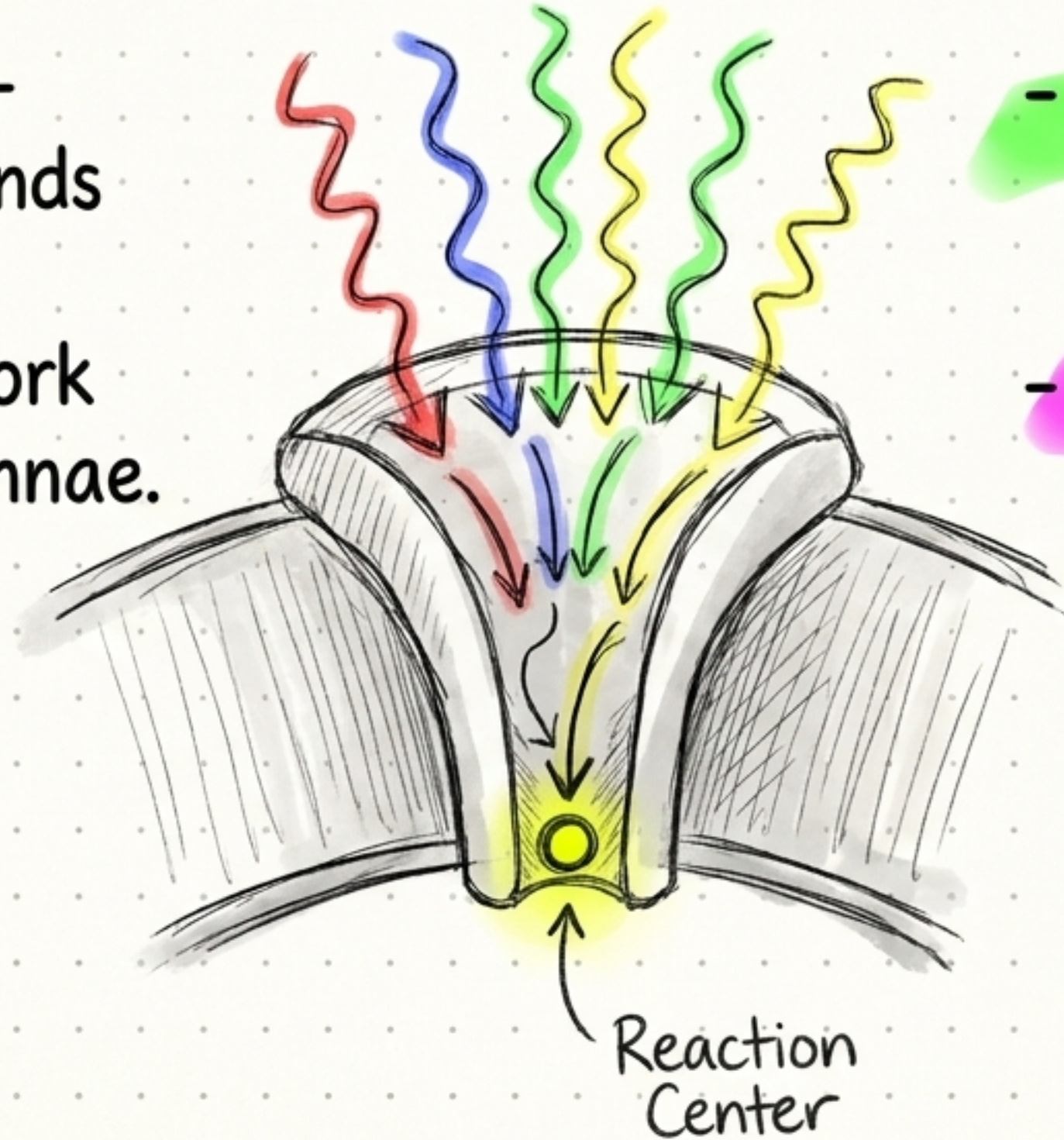


Stroma:

The fluid-filled space outside the thylakoids. This is where sugars are built.

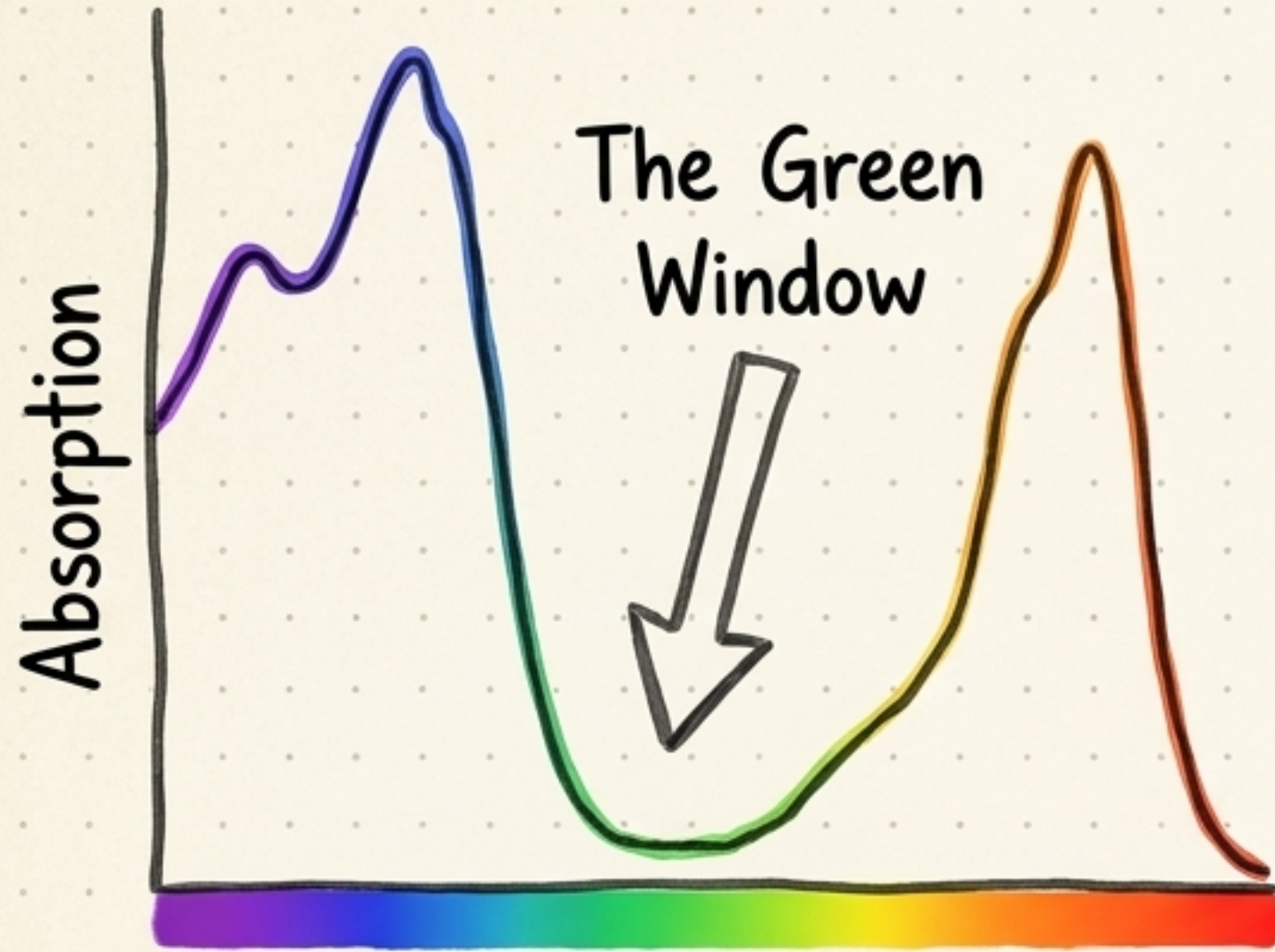
Catching the Light: The Pigment Bed

Pigments are light-absorbing compounds on the thylakoid membrane that work together like antennae.



- Chlorophyll a: The primary photosynthetic pigment.
- Accessory Pigments: Include Chlorophyll b, c, d, Carotenoids (orange/yellow), and Phycobiliproteins. They absorb wavelengths that Chlorophyll a misses and transfer the energy to the center.

Absorption Spectrum

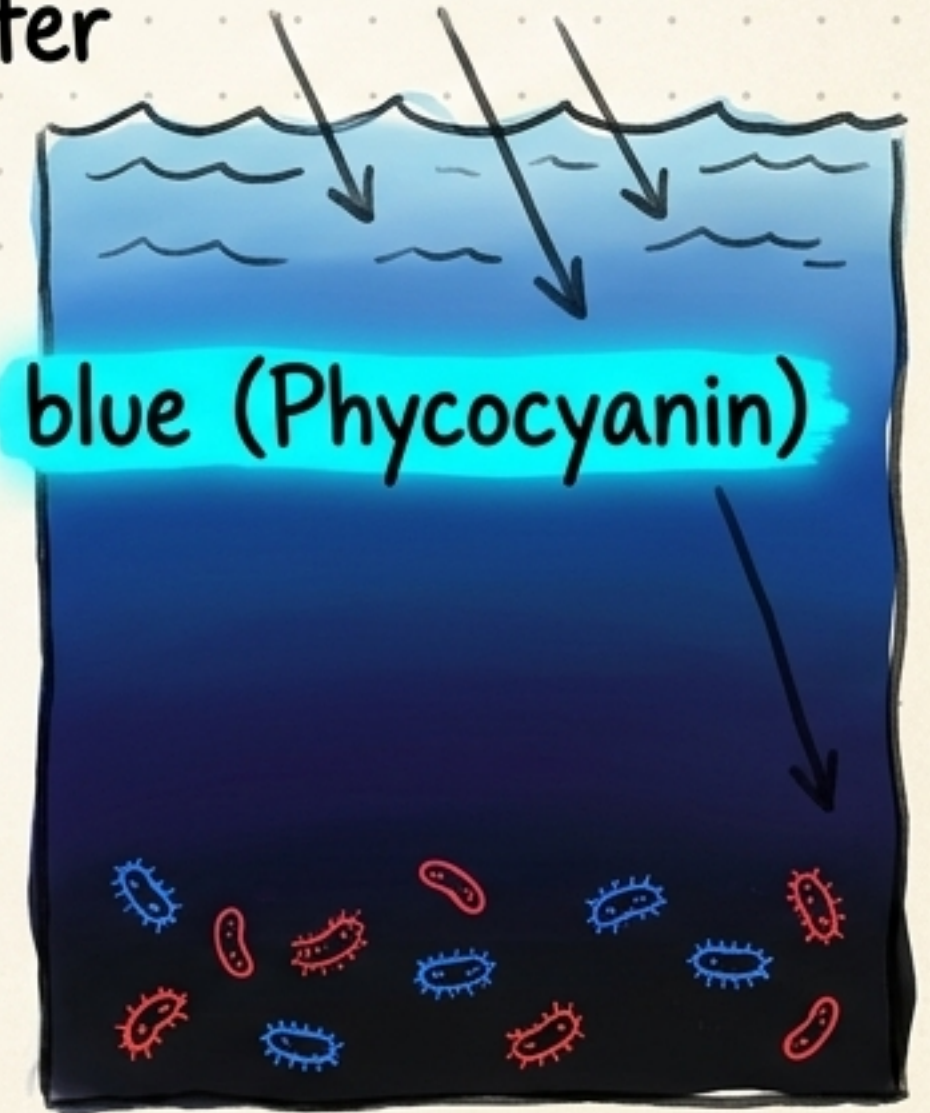


Plants appear green because chlorophyll reflects, rather than absorbs, green light.

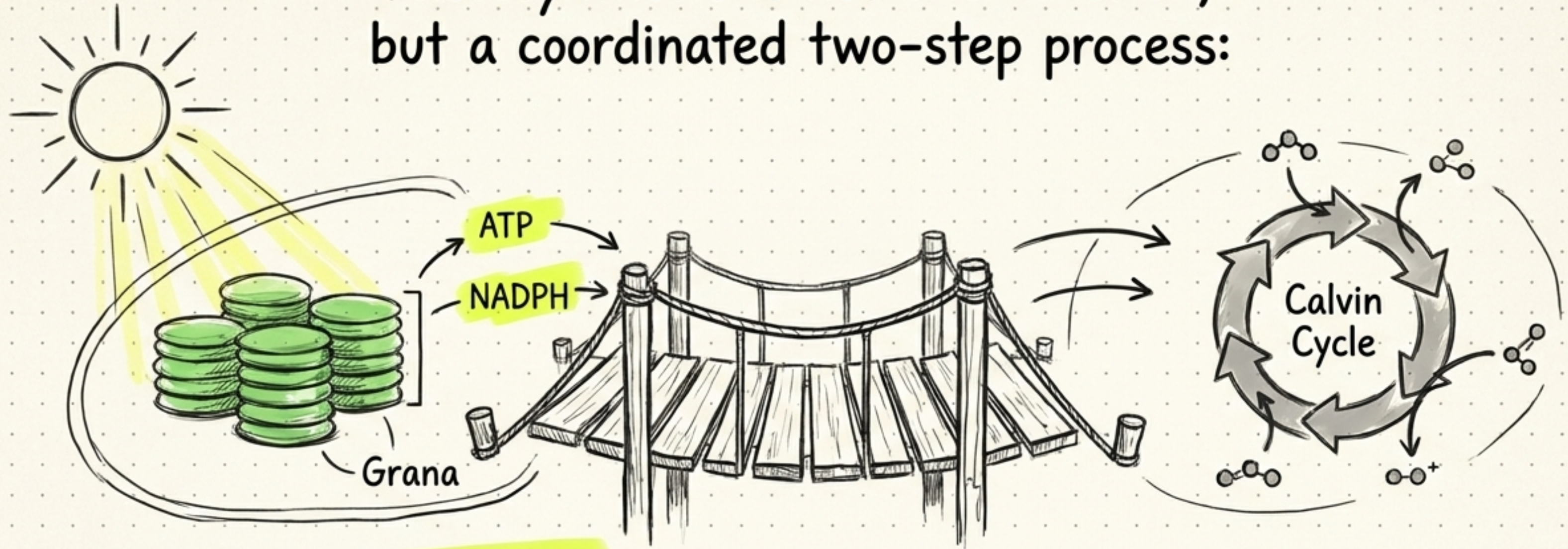
In aquatic ecosystems, water absorbs low frequencies (reds) while organic matter absorbs high frequencies (blues).

Organisms adapt their **Accessory pigments** to survive! Deep water cyanobacteria use use red

(Phycoerythrin) and blue (Phycocyanin) pigments to capture the exact wavelengths that penetrate the water.



Photosynthesis is not one reaction,
but a coordinated two-step process:



Act I: Light-Dependent Reactions (In the Grana)

- Captures solar energy to make chemical energy.

Act II: Light-Independent Reactions / Calvin Cycle (In the Stroma)

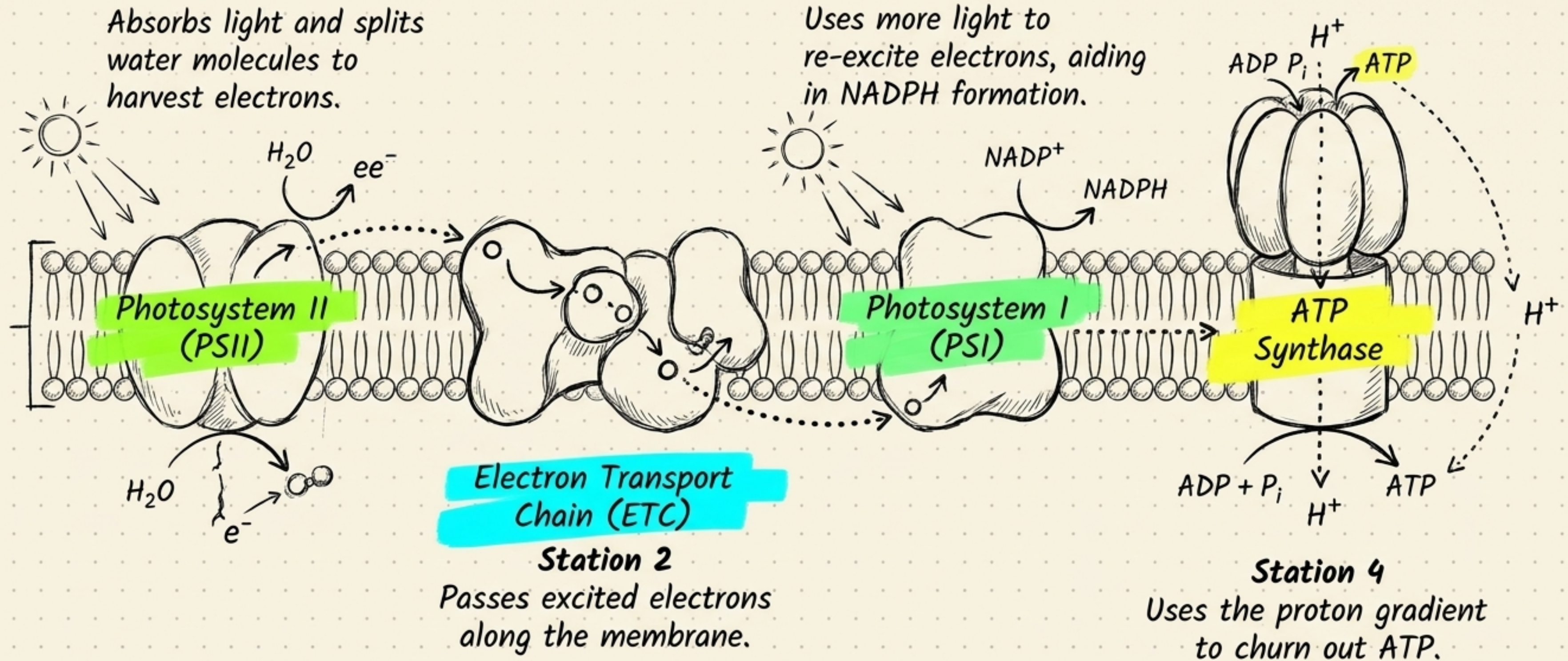
- Uses that chemical energy to build carbohydrates.

Act I: Light-Dependent Reactions



- Location: Thylakoid membranes.
- Goal: Capture solar energy and split water (Photolysis).
- The Byproduct: Oxygen is released (the most useful useless byproduct in history).
- The Output: Energy is trapped in two short-lived chemical carriers: ATP and NADPH.

The Machinery of Act I:



Act II: The Calvin Cycle (Dispelling the Myth)

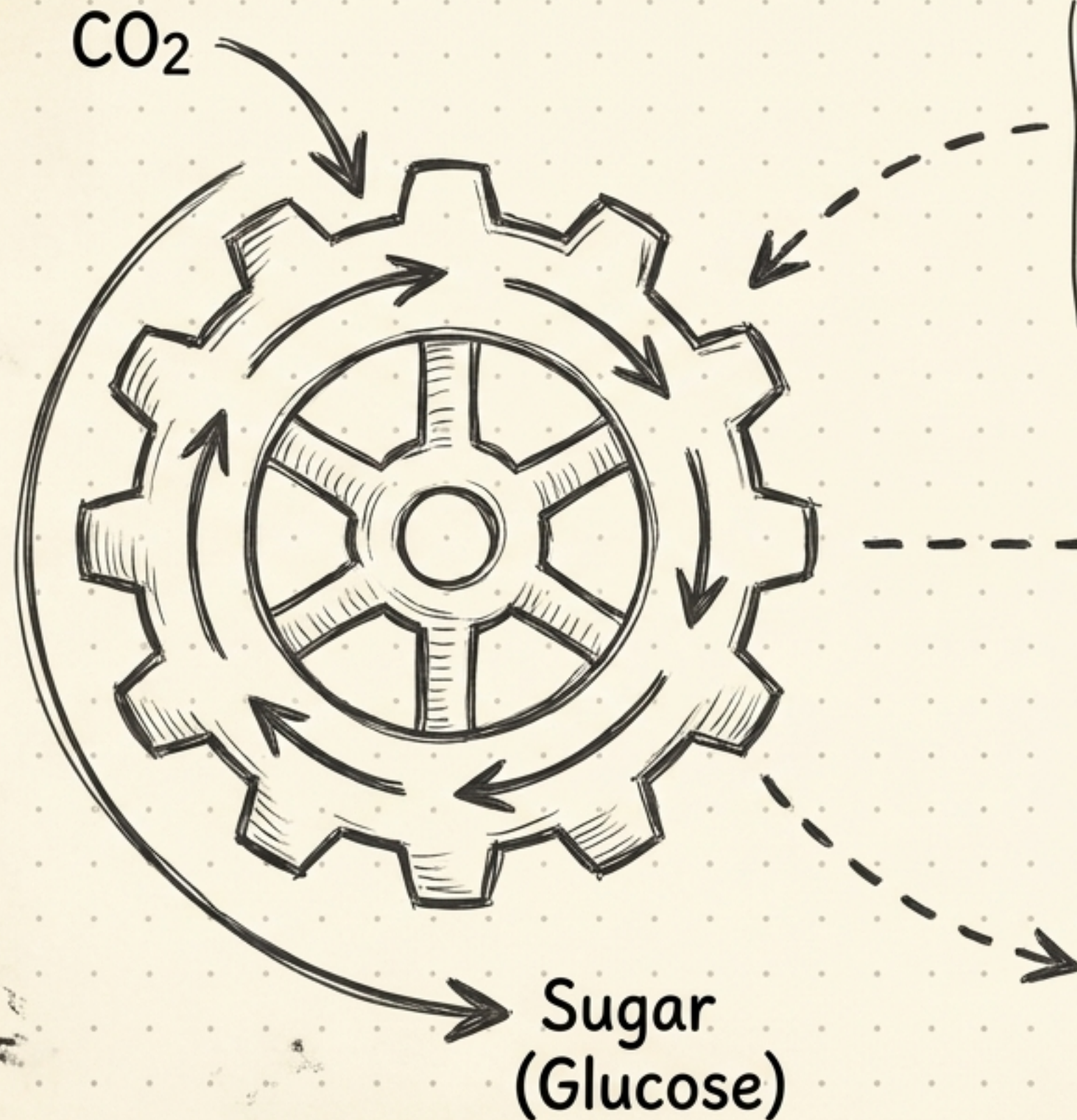


Often called the "~~Dark Reactions~~", but this is a lie!

The Calvin Cycle does NOT happen in the dark. It requires the short-lived **ATP** and **NADPH** produced by the light reactions, which quickly run out if the sun goes down.

Location: The **Stroma** (fluid inside the chloroplast).

The Assembly Line:

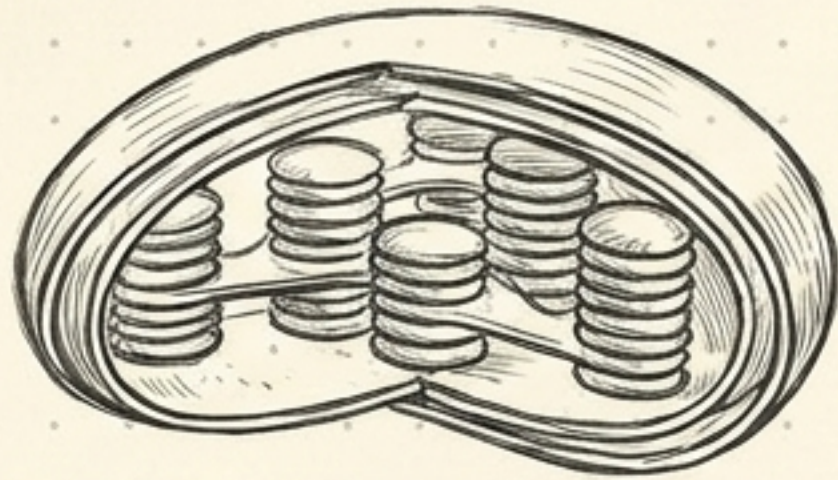


1. **Carbon Fixation:** The key enzyme **RuBisCO** grabs CO_2 from the air and attaches it to a 5-carbon molecule called RuBP.

2. Reduction: Energy from ATP and NADPH modifies these molecules into **G3P** (a simple sugar, half a glucose).

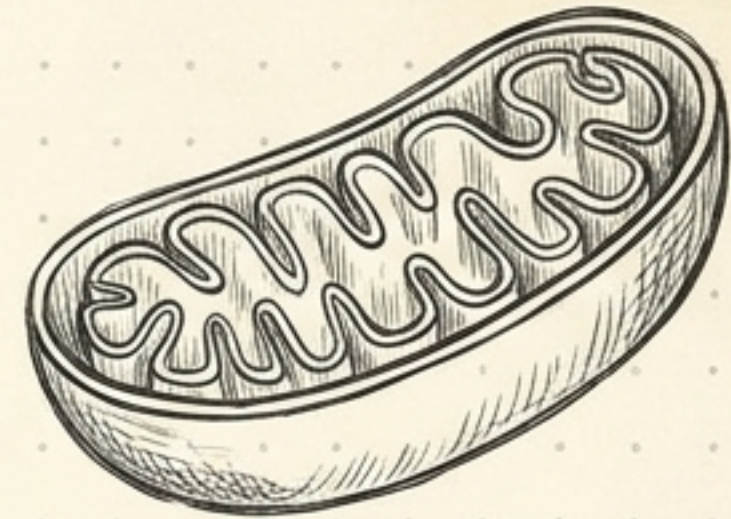
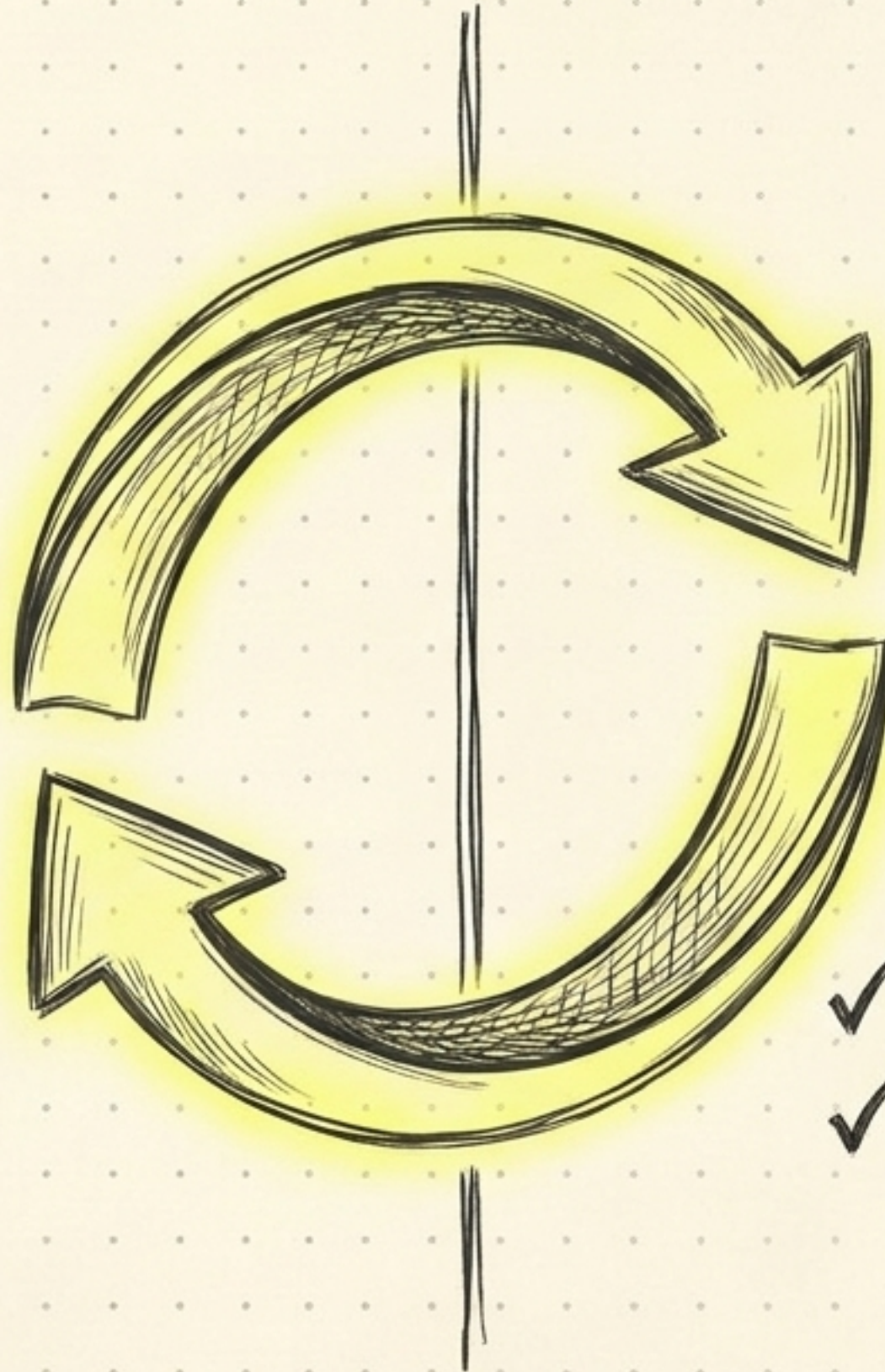
3. Regeneration: Some G3P exits to become glucose; the rest is recycled to rebuild RuBP, keeping the cycle turning!

If organelles could speak: The Cycle of Life



Photosynthesis (Chloroplasts):

- ✓ - Builds glucose from CO_2 (Anabolic)
- ✓ - Releases O_2
- ✓ - Occurs in plant cells & algae



Cellular Respiration (Mitochondria):

- Breaks down glucose (Catabolic)
- ✓ - Uses O_2 / Releases CO_2
- ✓ - Occurs in all Eukaryotes (Yes, plants have mitochondria too!)

The Whole Picture: Light energy powers the light-dependent stage to create chemical energy carriers, which power the Calvin Cycle to fix carbon into sugar.

